

IN5340/IN9340 Project I: Random Variables and Random Processes

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Agenda

1. Random Variables

- 1.1 Uniform random variable $X \sim \mathcal{U}(0, 1)$
- 1.2 Gaussian random variable $X \sim \mathcal{N}(\mu, \sigma^2)$
- 1.3 Central Limit Theorem

2. Stationarity and Ergodicity

Uniform RV: Task

- Generate a long sequence $x[n]$ of i.i.d. samples from $\mathcal{U}(0, 1)$.
- Estimate the PDF using a manually-normalized histogram.
- Estimate mean $m_x = \mathbb{E}[X]$ and variance $\sigma_x^2 = \mathbb{E}[(X - m_x)^2]$.
- Overlay the theoretical PDF and compare theory vs estimates.

Simulation in notebook: $N = 5000$ samples (seeded).

Uniform RV: Theoretical mean and variance (derivation)

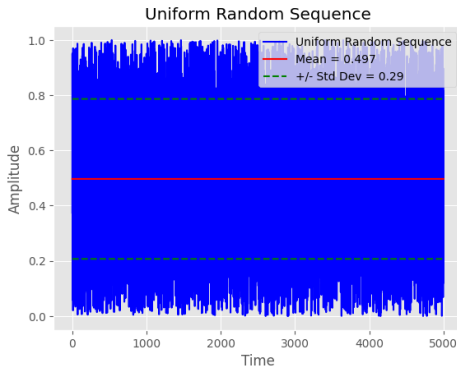
Assume $f_X(x) = 1$ for $x \in [0, 1]$ and 0 otherwise.

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^1 x dx = \left. \frac{x^2}{2} \right|_0^1 = \frac{1}{2}.$$

$$\mathbb{E}[X^2] = \int_0^1 x^2 dx = \left. \frac{x^3}{3} \right|_0^1 = \frac{1}{3}.$$

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{12} \approx 0.0833.$$

Uniform RV: Sequence and basic statistics



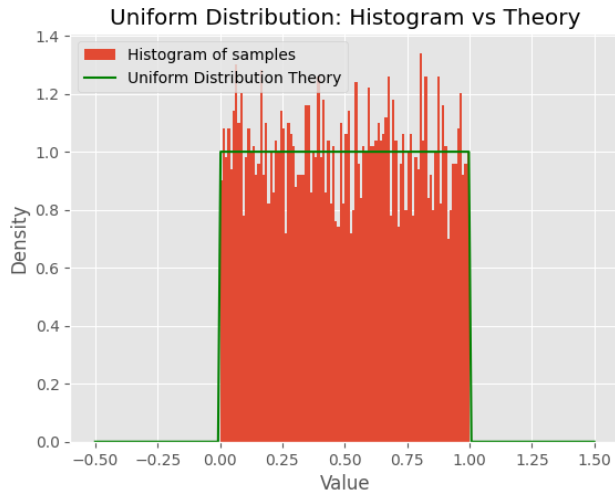
Estimated (one run):

- $\hat{m}_x \approx 0.4968$
- $\hat{\sigma}_x^2 \approx 0.08387$

Theory:

- $m_x = 1/2$
- $\sigma_x^2 = 1/12 \approx 0.08333$

Uniform RV: PDF estimate vs theory



Histogram is normalized to integrate to 1; the theoretical PDF is constant on $[0, 1]$.

Gaussian RV: Task

- Generate i.i.d. samples from a Gaussian RV (Matlab: `randn`, Python: `np.random.normal`).
- Estimate PDF (histogram) + mean and variance.
- Overlay the theoretical Gaussian PDF.

In the notebook: $\mu = 0$, $\sigma = 1$, $N = 5000$.

Gaussian RV: Theoretical mean and variance

For $X \sim \mathcal{N}(\mu, \sigma^2)$ with density

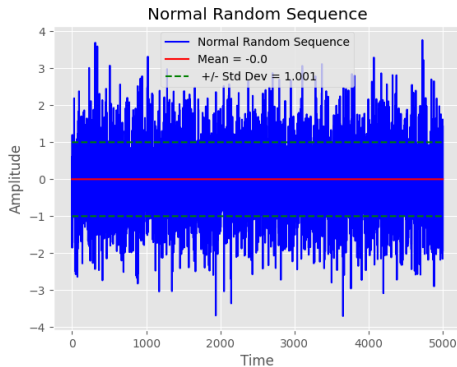
$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right),$$

the moments are

$$\mathbb{E}[X] = \mu, \quad \text{Var}(X) = \sigma^2.$$

Here: $\mu = 0$, $\sigma^2 = 1$.

Gaussian RV: Sequence



Estimated (one run):

- $\hat{\mu} \approx -2.2 \cdot 10^{-4}$

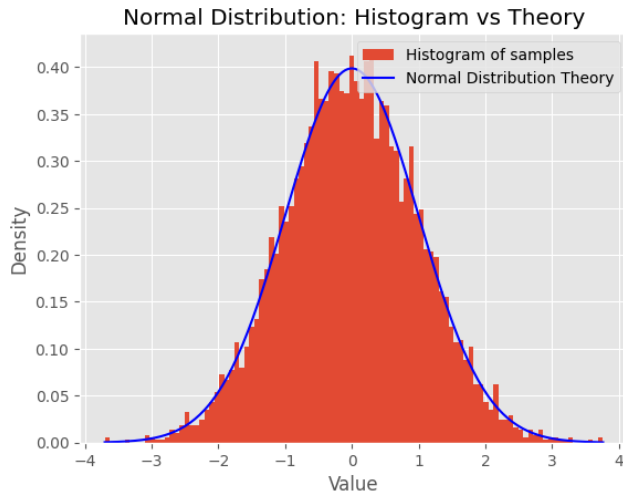
- $\widehat{\sigma^2} \approx 1.0013$

Theory:

- $\mu = 0$

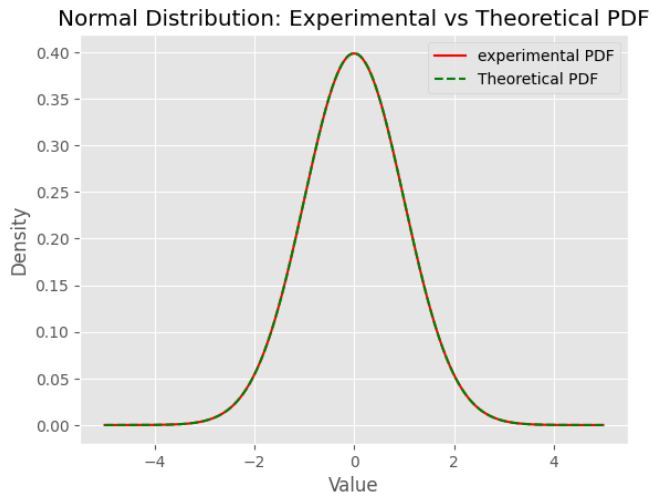
- $\sigma^2 = 1$

Gaussian RV: PDF estimate vs theory



Histogram-based density estimate with the theoretical Gaussian curve overlay.

Gaussian RV: Experimental vs theoretical PDF



- Generate 12 independent uniform RVs and average:

$$Y = \frac{1}{12} \sum_{k=1}^{12} X_k, \quad X_k \stackrel{iid}{\sim} \mathcal{U}(0, 1).$$

- Estimate the PDF of Y and compare to a Gaussian approximation.
- Derive theoretical mean and variance for Y .

CLT: Theoretical mean and variance (derivation)

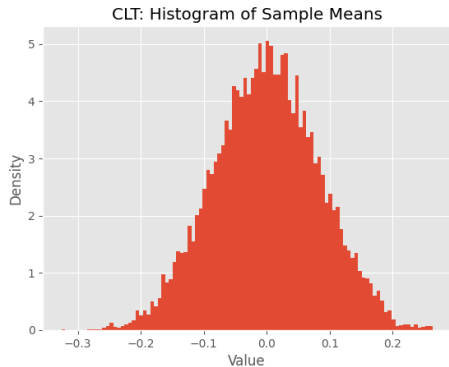
Using linearity of expectation and independence:

$$\mathbb{E}[Y] = \mathbb{E}\left[\frac{1}{12} \sum_{k=1}^{12} X_k\right] = \frac{1}{12} \sum_{k=1}^{12} \mathbb{E}[X_k] = \mathbb{E}[X] = \frac{1}{2}.$$

$$\begin{aligned}\text{Var}(Y) &= \text{Var}\left(\frac{1}{12} \sum_{k=1}^{12} X_k\right) = \frac{1}{12^2} \sum_{k=1}^{12} \text{Var}(X_k) \quad (\text{independent}) \\ &= \frac{12}{144} \cdot \frac{1}{12} = \frac{1}{144} \approx 0.00694.\end{aligned}$$

In the notebook, Y is centered as $Y_{\text{norm}} = Y - \mathbb{E}[Y]$, so $\mathbb{E}[Y_{\text{norm}}] = 0$ and $\text{Var}(Y_{\text{norm}}) = \text{Var}(Y)$.

CLT: Histogram of sample means



Estimated (one run):

- $\hat{m} \approx 0$ (after centering)
- $\hat{\sigma}^2 \approx 0.00693$

Theory:

- $\sigma^2 = 1/144 \approx 0.00694$

Random processes in the assignment

We analyze three random processes (generated from pseudo-random uniform noise):

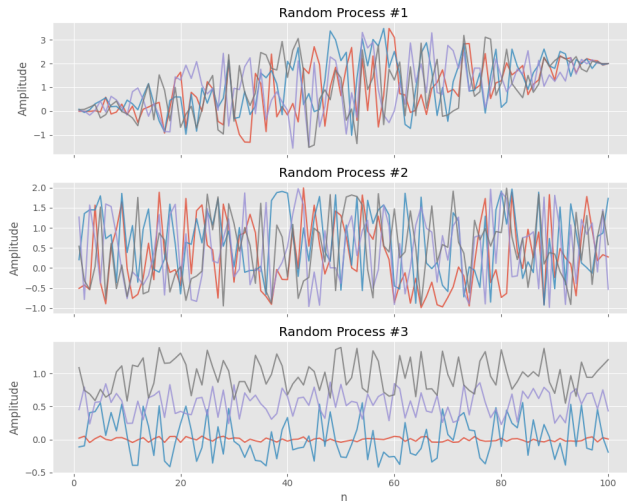
$$\text{rp1: } v[n] = (\text{rand} - 0.5) \underbrace{b \sin(\pi n/N)}_{\text{time-varying scale}} + \underbrace{an}_{\text{time-varying mean}},$$

$$\text{rp2: } v[n] = (\text{rand} - 0.5) m + a,$$

$$\text{rp3: } v[n] = (\text{rand} - 0.5) M_r + A_r, \quad A_r, M_r \text{ fixed per realization.}$$

Goal: decide stationarity/ergodicity by inspection and by ensemble/time averages.

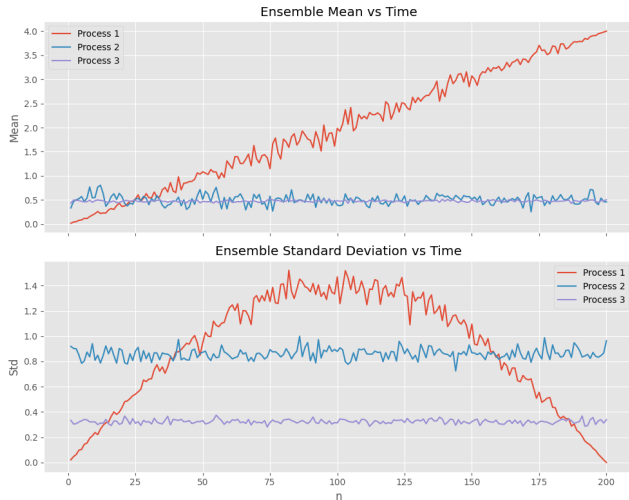
2.1 Inspection: realizations (M=4, N=100)



2.1 Stationary vs ergodic (qualitative conclusion)

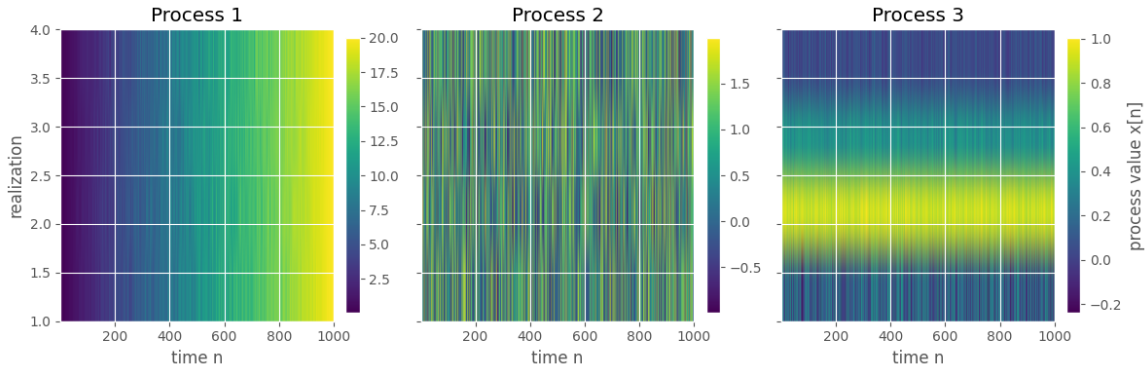
- **Process 1:** non-stationary (mean and scale vary with time) $\Rightarrow$ non-ergodic.
- **Process 2:** stationary (constant mean/variance) and appears ergodic in mean/variance.
- **Process 3:** stationary in distribution, but realization-dependent offset/scale $\Rightarrow$ non-ergodic.

2.2 Ensemble averages ($M=80$, $N=200$)



Ensemble mean/standard deviation vs time: rp1 varies with time; rp2 roughly constant; rp3 roughly constant but hides realization-dependent parameters.

2.3 Time averages + images ($M=4$, $N=1000$)



Imagesc-style view: time on x, realization on y. Time-variation across columns suggests non-stationarity; large realization-to-realization offsets suggest non-ergodicity.

2.3 Time averages: example output (one run)

- rp1 time-averages (per realization) are similar but the process is non-stationary, so not ergodic.
- rp2 time-averages cluster around the ensemble mean (consistent with ergodicity).
- rp3 time-averages differ significantly across realizations (non-ergodic in mean).

Example (from notebook stdout):

$$\bar{x}_{\text{time}}^{(rp2)} \approx [0.475, 0.453, 0.526, 0.515], \quad \bar{x}_{\text{time}}^{(rp3)} \approx [0.154, 0.909, 0.387, 0.047].$$

Summary

- Uniform and Gaussian RV experiments: estimated mean/variance and PDF match theory with small finite-sample deviations.
- CLT: the average of 12 uniforms is well-approximated by a Gaussian; theory gives $\sigma^2 = 1/144$.
- Random processes: rp1 non-stationary/non-ergodic; rp2 stationary/ergodic; rp3 stationary but non-ergodic.

All code and figures reproduced from the accompanying notebook.